Protein structure prediction by autoregressive modeling of residue contacts

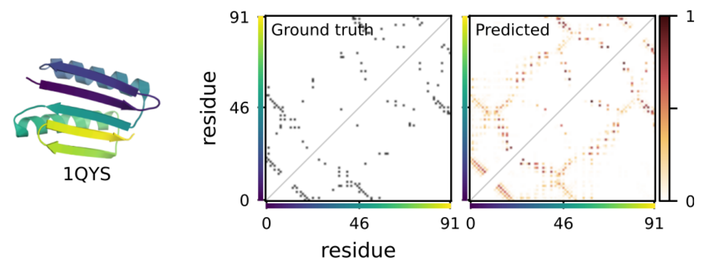
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Open Athena

Summary

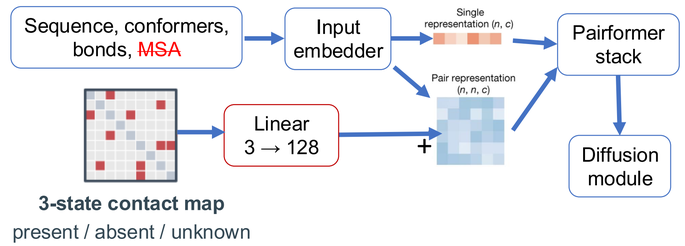
- Existing protein structure predictors rely on evolutionary information in the form of multiple sequence alignments (MSAs) or pre-trained protein language model (PLM) embeddings. Accuracy suffers in settings with less informative evolutionary information.
- **MarinFold** is a 1.5B-parameter decoder-only transformer trained from scratch to predict protein contact maps autoregressively without MSAs or pre-trained PLM embeddings.
- **Helico**, a fine-tuned Protenix model, converts the predicted contacts into 3D structures
- MarinFold can predict accurate contact maps for many proteins, but falls short of the accuracy of state-of-the-art predictors that explicitly use evolutionary information
- Surprisingly, accuracy remains correlated with effective MSA depth.
- We are exploring training data improvements, inference-time search, model scaling, post-training, and prediction of protein complexes

Example prediction: Top7 (1QYS)



Generating 3D structures

We fine-tuned Protenix-v1 [4], an AlphaFold 3-like model, to accept a contact map in place of an MSA. We call the resulting model Helico.



Training MarinFold on predicted structures

Train on structures predicted by AlphaFold2 or ESMFold2
Evaluate on a 314-protein subset of FoldBench [5]

Data source	Training examples
ESM-Atlas [1]	66 million
AlphaFold DB [2]	4 million
Redesigned sequences for above structures (ProteinMPNN) [3]	163 million



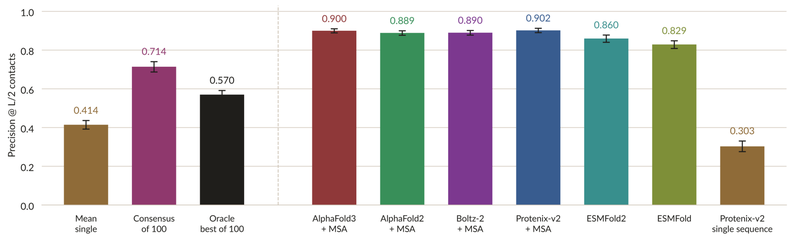
Contact maps

GPT-style language model
Trained from scratch
No natural language data
Decoder only (Qwen3 architecture) [11]
Marin infrastructure (GPUs, TPUs) [12]
1.5B parameters
Context length: 8,192

```
<contacts-v1> <begin_sequence>
<p28> <VAL> <p43> <ARG>  …
<begin_statements>
<contact> <p46> <p67>
<contact> <p6> <p41>
<contact> <p47> <p83>
<contact> <p87> <p45>
```

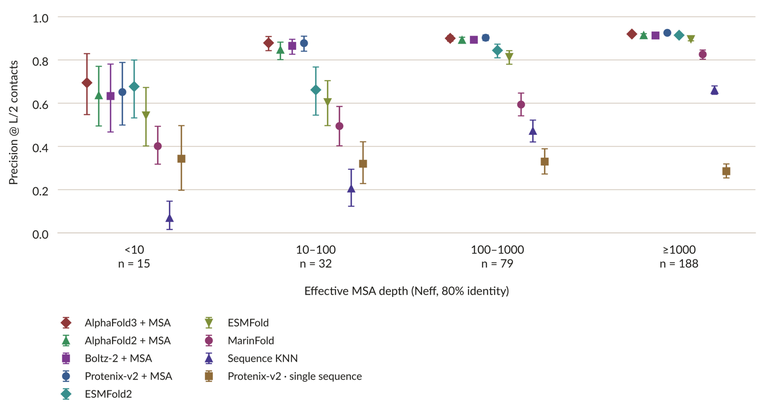
Training documents
232M documents, 249B tokens
Each <word> is a token

Contact prediction on FoldBench monomers

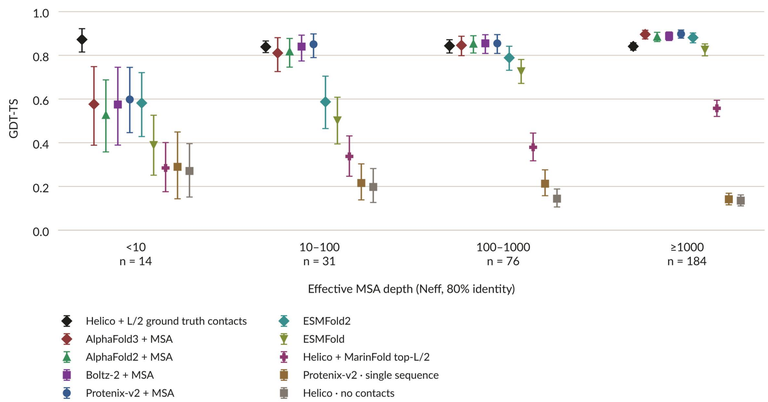


Precision among the top L/2 contacts (L = sequence length), n = 314.
Training proteins with >=30% sequence identity to any eval protein were removed prior to training.

Accuracy correlates with effective MSA depth



Structure prediction accuracy



Conclusions and ongoing work

MarinFold + Helico outperforms Protenix v2 in single sequence mode overall, but falls short of stronger baselines.

We are exploring:
- Understanding what MarinFold learned
- Improving accuracy on low MSA depth proteins through training data composition
- Applying inference-time search and post-training
- Expansion beyond protein monomers to protein/protein complexes or other systems

MarinFold is openly developed. Contributions welcome.

github.com/Open-Athena/MarinFold

References

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